

**A Web and Mobile-Based Point-of-Sales System with
Email Notification for Mr. Siklo Motor Parts and Gears Trading**

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Chapter 1

Introduction

1.1 Background of the Study

In the modern era, businesses rely heavily on technology to streamline their operations and enhance customer satisfaction. Inventory management is a crucial aspect of any business, as it ensures that the right products are available at the right time. Mr. Siklo Motor Parts and Gears Trading is a small business that specializes in the sale of motor parts and accessories. Despite the owner's extensive knowledge of motor parts, the business faces significant challenges in managing inventory and processing sales effectively.

Currently, the business operates with a computerized process where sales are recorded using Microsoft Excel. While this provides a basic structure for tracking transactions, it still leaves room for inefficiencies and errors. Due to great quantity of products offered, the owner struggles with determining which products need restocking, often relying on manual counts and educated guesses. This situation sometimes leads to stockouts, negatively affecting customer satisfaction and potential sales.

The advent of web-based systems presents an opportunity to address these issues by automating inventory and sales management. The researcher aims to develop a web and mobile-based ordering, sales and inventory management system for Mr. Siklo Motor Parts and Gears Trading, which will streamline operations by enabling users to effortlessly browse products, add items to their cart, and complete purchases through convenient e-payment options such as Gcash. Additionally, the system will feature email notifications to keep customers informed about their order statuses, thereby improving customer engagement and satisfaction. This research seeks to address existing challenges in inventory management and sales efficiency while providing a user-friendly experience.

1.2 Problem and Objectives of the Study

1.2.1 General Problem

Mr. Siklo Motor Parts and Gears Trading faces difficulties in managing its inventory and tracking sales, leading to inefficiencies in restocking and order processing.

1.2.2 Specific Problem

- The ordering process is done manually on paper, leading to inefficiencies and delays in fulfilling customer orders.
- Inventory management is done manually through Excel, causing inaccurate stock tracking and difficulties in determining which products need to be restocked in a timely manner.
- Sales transaction records are inputted manually through Excel, increasing the risk of human error and affecting the accuracy of sales data.
- Customers do not receive real-time updates regarding the status of their orders, leading to a lack of transparency and decreased customer satisfaction.

1.2.3 General Objective

The general objective of this study is to design and develop a web and mobile-based ordering, sales and inventory management system for Mr. Siklo Motor Parts and Gears Trading that automates sales tracking and inventory management while providing reminders for stock levels and notifications for order statuses via email.

1.2.4 Specific Objectives

- To automate the ordering process, enabling faster and more efficient order fulfillment.
- To develop a real-time inventory management system that ensures accurate stock tracking and alerts the owner when products are low on stock.
- To create an automated sales tracking system that records transactions accurately and reduces the risk of human errors.
- To implement an email notification system that keeps customers informed about their order status, from confirmation to delivery.

1.3 Scope and Limitation of the Study

1.3.1 Scope of the Study

- This study focuses on the development of a web and mobile-based ordering, sales and inventory management system for Mr. Siklo Motor Parts and Gears Trading.
- This study will develop a user interface to enable interactions with the product catalog, cart functionalities, and the checkout process on both desktop and mobile platforms.
- This study will provide inventory tracking by automatically updating stock levels based on sales and prompting low stock alerts to the owner.
- The study will support payment through digital platforms, specifically GCash, allowing customers to pay conveniently online.
- This study will allow to receive real-time updates on the status of their purchases through email notifications.
- The study will also include generating sales and inventory reports to help the business make data-driven decisions.

1.3.2 Limitations of the Study

- This study is limited to sales and inventory management processes and does not extend to other business functions, such as payroll, taxes, or complex financial accounting.
- The study utilizes PHPMailer for handling email notifications directly within the system.
- The implementation of the desktop web platform will include both the user side and the admin side.
- The implementation of the mobile web platform will be limited to the user side.

1.4 Significance of the Study

Several stakeholders who are associated with the operations of Mr. Siklo Motor Parts and Gears Trading stand to gain from this research. Specifically, the following stakeholders shall benefit from the study:

Mr. Siklo Motor Parts and Gears Trading Management. The owner and management will directly benefit from a web-based sales and inventory management system. This study will be able to provide real-time sales and inventory data, enabling better decision-making, improved stock management, and reduced operational costs. Accurate reports will help forecast demand, preventing stockouts or overstocking, and boosting business growth.

Mr. Siklo's Customers. The study will enhance customer experience by allowing them to browse and order products from the comfort of their homes, improving convenience. Additionally, timely order updates and ensured product availability will result in faster service and greater satisfaction.

Staff. Mr. Siklo Motor Parts and Gears Trading employees will experience reduced workloads with automated sales and inventory tracking, allowing them to focus more on

customer service and reducing the risk of errors in inputting data through their current computerized processes.

Suppliers. With more accurate inventory management, Mr. Siklo Motor Parts and Gears Trading will be able to maintain better relationships with suppliers. It will reduce the likelihood of supply shortages and last-minute orders, promoting more efficient and reliable supply chain operations.

Motors Parts and Gears Industry. This study can serve as a reference for similar small businesses, demonstrating how the integration of technology can optimize operations, improve efficiency, and enhance overall business performance.

Researchers. The researchers conducting this study will gain valuable insights into inventory and sales management systems, enhancing their understanding of business process automation. This experience will contribute to their academic and professional development, providing practical applications of their knowledge.

Future Researchers. This research contributes to the field of business process automation, providing a foundation for future studies on inventory and sales management systems. It can inspire further exploration into technology integration in small businesses and its impact on operational efficiency.

1.5 Definition of Terms

Inventory Management – The process of overseeing and controlling the ordering, storage, and use of products or materials within a business.

Web-Based System – A system that is accessible via the internet and runs on web browsers rather than being installed on individual computers.

Mobile-Based System – A system that operates on mobile devices, allowing users to access applications and services for tasks like ordering management.

Stock Level – The quantity of a product or material that is available for sale in a business.

Notification System – A feature that alerts users through email about order status updates.

Sales Tracking – The process of recording and monitoring sales transactions within a business.

Chapter 2

Conceptual Framework

The conceptual framework for the study "A Web and Mobile-Based Point-of-Sales System with Email Notification for Mr. Siklo Motor Parts and Gears Trading" illustrates the relationship between the system's essential components and the business's operational workflows. The framework integrates technical and operational aspects to streamline processes, improve efficiency, and enhance customer satisfaction.

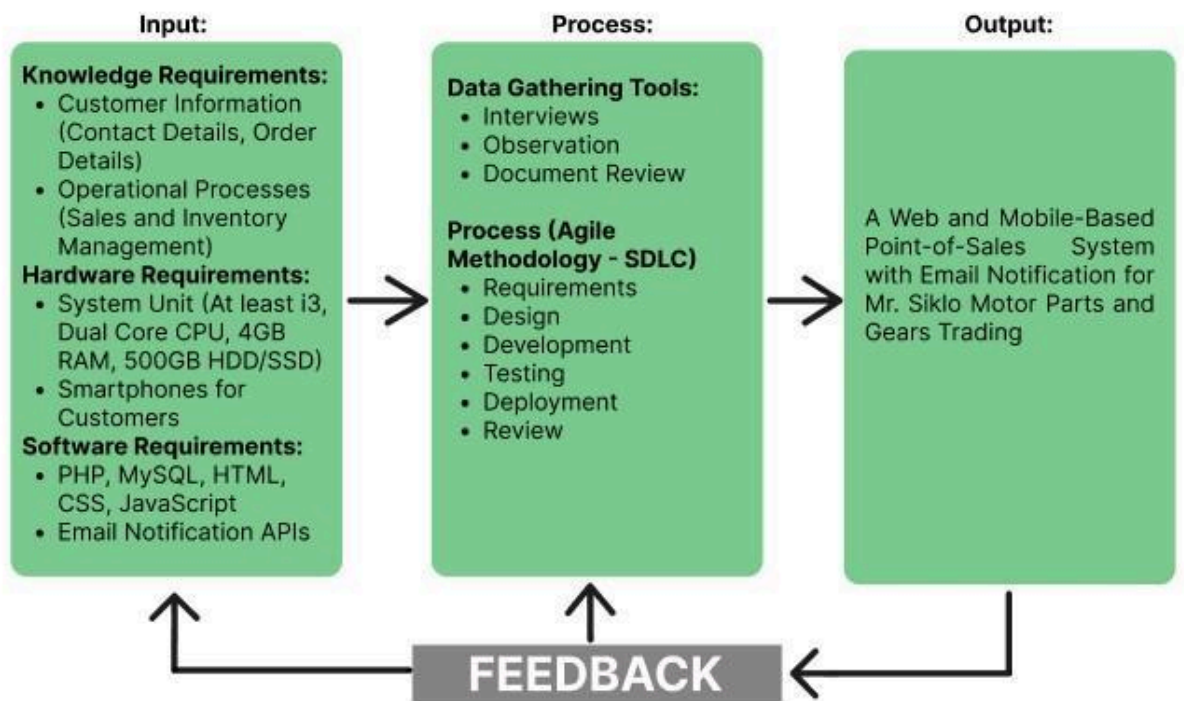


Figure 1. Conceptual Framework

The conceptual framework illustrates the flow of data, processes, and outcomes for the proposed "Web and Mobile-Based Point-of-Sales System with Email Notification for Mr. Siklo Motor Parts and Gears Trading." This framework serves as a guide to understanding how the system will address the challenges faced by the business, such as inefficient sales tracking, inaccurate inventory management, and the lack of timely updates for customers.

The process begins with the Input Phase, where essential requirements for the system are gathered. These requirements are categorized into three components: knowledge requirements, hardware requirements, and software requirements. Knowledge requirements include customer information, product inventory data, and sales transaction records, which are critical for ensuring accurate system functionality. Hardware requirements, such as system units with specifications (at least i3 processors, dual-core

CPU, 4GB RAM, and 500GB HDD/SSD), along with smartphones for customers, ensure that both the admin and user sides of the system can operate efficiently. Software requirements involve programming languages and tools like PHP, MySQL, HTML, CSS, and JavaScript, as well as email notification APIs for automated communication. These inputs collectively form the foundation of the system, ensuring it is built with the right resources and data to address the operational needs of the business.

The Process Phase is divided into two main parts: data gathering tools and the system development process. Data gathering tools include interviews, observations, and document reviews, which allow the researchers to understand the current challenges faced by the business and the requirements for the proposed solution. The system development process follows the Agile Software Development Life Cycle (SDLC), ensuring the system is created in a structured and iterative manner. The SDLC includes several key stages: requirements gathering, system design, development, testing, deployment, and review. During the requirements stage, the system's functionalities are clearly defined based on the business needs. In the design stage, user-friendly interfaces for both the admin and customer sides are developed. The development stage involves coding the core features, such as automated sales tracking, real-time inventory updates, stock alerts, and email notifications. Testing ensures that the system is free from errors and meets the desired specifications. Deployment involves implementing the system in the business environment, while the review ensure that the system is ready for full use and aligned with the goals of the business.

The Output Phase highlights the end result of the entire process: A Web and Mobile-Based Point-of-Sales System tailored to the needs of Mr. Siklo Motor Parts and Gears Trading. The system features automated sales tracking to eliminate errors in recording transactions, real-time inventory management to monitor stock levels accurately, and low stock alerts to notify the business owner when products need restocking. It also includes digital payment integration, such as GCash, for seamless

transactions, and an email notification system to update customers on the status of their orders. These outputs aim to address the existing challenges in managing sales and inventory, improving operational efficiency, and enhancing customer satisfaction by providing timely and transparent communication.

A crucial aspect of the conceptual framework is the Feedback Loop, which ensures continuous improvement of the system. Insights gained from the output phase—such as user feedback, performance data, and system usage patterns—are fed back into the input phase. For example, sales reports and inventory data can help refine stock management practices, while user feedback can guide updates to the system's interface or features. This iterative feedback mechanism ensures that the system evolves over time to meet the changing needs of the business, providing a sustainable and long-term solution.

In conclusion, the conceptual framework demonstrates a seamless integration of input, process, and output phases, supported by a feedback loop for ongoing optimization. By incorporating advanced features such as automated processes, realtime updates, and email notifications, the proposed system aims to improve business operations, enhance decision-making, and deliver a better experience for both the business owner and customers of Mr. Siklo Motor Parts and Gears Trading.

Review of Related Literature

In this phase, the researchers have reviewed the relevant literature and research to understand the proposed topic.

Local Literature

Inventory Management Tools for Local Micro Businesses

The study conducted by Jeremiah Cordial (2020) explores inventory management tools specifically tailored for local micro businesses in the Philippines, such as sari-sari stores and carinderias. The research emphasizes the significance of implementing efficient inventory management systems to enhance operational effectiveness and profitability for these small enterprises. It identifies various inventory management techniques, including just-in-time (JIT), safety stock, first-in-first-out (FIFO), batch tracking, and demand forecasting, which can be integrated into the daily operations of micro businesses. The study also highlights the necessity of understanding inventory management to reduce costs, prevent stockouts, and improve customer satisfaction, ultimately contributing to the competitiveness of micro businesses in the local market.

Retrieved from:

https://www.academia.edu/51805887/Inventory_Management_Tools_for_Local_Micro_Businesses

The findings from Cordial's study resonate with the objectives of developing a web and mobile-based ordering, sales, and inventory management system for Mr. Siklo Motor Parts and Gears Trading. Just as the study identifies the need for micro businesses to adopt inventory management tools to optimize their operations, Mr. Siklo aims to automate sales tracking and inventory management to alleviate the challenges faced by its owner in managing stock levels in Excel. By incorporating features such as real-time inventory updates and Email notifications for order statuses, Mr. Siklo can enhance customer engagement and satisfaction, paralleling the benefits outlined in the research. This alignment underscores the importance of integrating technology in micro business operations to achieve greater efficiency and customer service excellence.

Cordial, J. B. (2020). *Inventory Management Tools for Local Micro Businesses*. Bicol University.

Nexus between payments digitalization and cash usage in the Philippines

The study by Glindro et al. (2024) examines the relationship between digital payments and cash usage in the Philippines, finding that while digital payment platforms are gaining increased acceptance, the substitution from cash to digital payments remains relatively small. Their research indicates that despite significant progress in payment digitalization initiatives, there remains substantial opportunity to expand digital payment adoption. The authors emphasize that the pace and extent of digital payment adoption depends heavily on policy reforms aimed at widening access through affordable services, secure infrastructure, and strong privacy protection measures.

Retrieved from: <https://pre.econ.upd.edu.ph/index.php/pre/article/view/1049>

This research relates significantly to the Mr. Siklo Motor Parts and Gears Trading study as it provides valuable insights into the current state of digital payments adoption in the Philippines. As the proposed system for Mr. Siklo includes GCash integration for online payments, understanding the broader context of digital payment adoption and its challenges helps inform implementation strategies. The findings from Glindro et al. (2024) suggest that while implementing digital payment options is important for modernization, the system should maintain flexibility to accommodate both digital and traditional payment methods, ensuring accessibility for all customer segments. This aligns with the study's objective to enhance customer convenience while acknowledging potential limitations in digital payment adoption.

Glindro, E. T., Ofiaza, R. E., & Jose, M. K. Q. (2024). Nexus between payments digitalization and cash usage in the Philippines. *The Philippine Review of Economics*, 61(1), 44-70.

Dynamic relationships: E-commerce sales and key exogenous variables in the Philippines

The study conducted by Montano and Mercado (2023) explores the intricate dynamics of e-commerce sales in the Philippines, identifying key exogenous variables such as digital marketing spending, GDP growth, internet penetration, and mobile phone ownership that significantly influence sales outcomes. Utilizing a flexible spline modeling approach, the research uncovers non-linear relationships among these variables, highlighting the critical role of digital marketing in driving e-commerce growth. The findings indicate that as the Philippines continues to expand economically and integrate technologically, these factors will increasingly shape the e-commerce landscape, presenting both opportunities and challenges for businesses and policymakers alike (Montano & Mercado, 2023).

Retrieved from: <https://oapub.org/soc/index.php/EJMMS/article/view/1603>

This research is directly relevant to our study on Mr. Siklo Motor Parts and Gears Trading, as it underscores the importance of digital marketing and technology in enhancing sales and inventory management within the e-commerce sector. By implementing a web and mobile-based ordering system, Mr. Siklo can leverage insights from the study to optimize inventory management and improve customer engagement through timely notifications and efficient sales tracking. Furthermore, understanding the impact of internet penetration and mobile phone ownership on consumer behavior will help Mr. Siklo tailor his marketing strategies to effectively reach and serve a growing online customer base. Overall, the findings from Montano and Mercado's study provide a valuable framework for enhancing operational efficiency and customer satisfaction in Mr. Siklo's business.

Montano, V. E., & Mercado, J. L. L. (2023). Dynamic relationships: E-commerce sales and key exogenous variables in the Philippines. *European Journal of Management and Marketing Studies*, 8(3), 111-130.

Why a POS system is important in retail business

According to Luna (2021), A POS system is an essential tool for modern retail businesses, offering numerous benefits that streamline operations and improve customer satisfaction. It optimizes the checkout process through faster transactions, supports various payment channels like e-wallets and credit cards, and enhances inventory management by automating stock tracking and minimizing errors. Additionally, a POS system provides accurate data for better decision-making, such as identifying top-selling products and managing promotions, while cloud-based systems enable remote access to monitor operations anytime. By adopting a POS system, businesses can enhance efficiency, reduce errors, and stay competitive in a rapidly evolving retail landscape.

Retrieved from:

https://mb.com.ph/2021/10/08/why-a-pos-system-is-important-in-retail-business/#google_vignette

In relation to Mr. Siklo Motor Parts and Gears Trading, a POS system is essential as it directly addresses the challenges the business faces in managing orders, inventory, and sales. By automating the checkout process, the system enables faster and more accurate transactions, improving efficiency and reducing delays. It also streamlines inventory management by automatically updating stock levels and providing low-stock alerts, ensuring that the business avoids stockouts or overstocking. Additionally, the integration of modern payment methods, such as e-wallets like GCash, aligns with customer preferences and enhances convenience. The POS system's ability to generate accurate sales and inventory reports will help Mr. Siklo make informed decisions about restocking, promotions, and overall business strategy. By incorporating features such as email notifications and a user-friendly interface, the system will not only improve operations but also enhance customer satisfaction, making it a vital component of the proposed study.

Luna, E. (2021, October 8). Why a POS system is important in retail business. Manila Bulletin.

Web-Based Abacus Ship Management Asset Tracking and Order Processing with Decision Support System

According to Ladia et al. (2023), the development of an Asset Tracking and Order Processing with Decision Support System for ABACUS Ship Management Ltd., Manila, effectively automated manual processes such as inventory management, order tracking, and generating annual historical reports. The system allowed multiple users to track orders, manage inventory levels, and select supplier quotations while providing comprehensive and accurate reports. Additionally, it was found to be reliable, portable, user-friendly, and efficient, enhancing the company's operations. To further improve the system, the researchers recommended features like SMS notifications, a chat system for better communication, automated Request for Quotation (RFQ) functionalities, and porting the system to a mobile application for increased accessibility.

Retrieved from: <https://ejournals.ph/article.php?id=19657>

The study on the Web-Based Abacus Ship Management Asset Tracking and Order Processing with Decision Support System by Ladia et al. (2023) relates to Mr. Siklo's study as both aim to automate and optimize manual processes in inventory management, order tracking, and reporting. Similar to the Abacus system, Mr. Siklo's proposed system seeks to streamline operations by automating stock level monitoring, tracking sales and orders, and generating detailed reports to aid decision-making. Both systems emphasize improving efficiency, user-friendliness, and reliability, addressing the challenges of managing inventory and orders in a manual setup. Additionally, features such as notifications and mobile accessibility suggested in the Abacus study could serve as valuable insights for enhancing Mr. Siklo's system, ensuring it not only meets current business needs but also supports future growth and technological advancements.

Ladia, J. M., Cuaresma, S. A., & Deynata, D. R. (2023). Web-Based Abacus Ship Management Asset Tracking and Order Processing with Decision Support System. <https://ejournals.ph/article.php?id=19657>

Foreign Literature

Spare Parts Inventory Management: A Literature Review

According to Zhang, Huang, and Yuan (2021), spare parts inventory management is a critical aspect of supply chain operations, particularly for maintaining capital goods and consumer durable products. The research highlights various challenges in this field, including the need for efficient inventory policies, demand forecasting, and life-cycle planning. It emphasizes the differences between managing spare parts for capital goods versus consumer durables, noting that the latter often requires a cost-minimizing approach due to shorter product life cycles and rapid technological changes. The study identifies several research gaps, such as the lack of life-cycle planning perspective in consumer durable goods spare parts management and the need for more advanced warranty contract considerations. Additionally, it points out the potential benefits of implementing big data analytics and exploring the integration of spare parts management with concepts like reverse logistics and supply chain sustainability.

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[27/publication/349654272_Spare_Parts_Inventory_Management_A_Literature_Review/links/60be735992851cb13d88b987/Spare-Parts-Inventory-Management-A-Literature-Review.pdf](https://www.researchgate.net/profile/Shuai-Zhang-27/publication/349654272_Spare_Parts_Inventory_Management_A_Literature_Review/links/60be735992851cb13d88b987/Spare-Parts-Inventory-Management-A-Literature-Review.pdf)

In relation to Mr. Siklo Motor Parts and Gears Trading, the insights from this literature review are directly relevant to the proposed web and mobile-based ordering, sales, and inventory management system. The emphasis on efficient inventory policies aligns with the project's goal of automating stock level tracking and implementing reorder alerts, while the need for accurate demand forecasting resonates with the system's aim to generate detailed sales and inventory reports to support informed decision-making. Although Mr. Siklo's business focuses on consumer motor parts rather than capital goods, the challenges of managing spare parts for consumer durables discussed in the study are particularly applicable. The integration of e-commerce features and email notifications in the proposed system addresses the need for improved customer service and communication, which are critical in the spare parts industry. Additionally, the study's

mention of big data analytics highlights the potential for leveraging data collected through the new system to enable advanced analytics in the future, further enhancing inventory management and customer satisfaction.

Zhang, S., Huang, K., & Yuan, Y. (2021). Spare parts inventory management: A literature review. *Sustainability*, 13(5), 2460.

Inventory Management Optimization for Intermittent Demand

The study by Kaya et al. (2024) addresses the challenges faced by an electrical equipment company in managing inventory and forecasting demand. The company struggles with intermittent demand patterns due to inadequate analytical methodologies, leading to issues such as excess inventory and customer dissatisfaction. To tackle these challenges, the authors developed an inventory management system that integrates the Newsvendor and Order Up Policy, utilizing the Syntetos-Boylan Approximation (SBA) and Croston methods for demand forecasting. A mathematical model was created to optimize order levels while considering lead times and costs, ultimately aiming to minimize expenses and enhance customer satisfaction. The study emphasizes the importance of accurate demand forecasting and presents a comprehensive framework for inventory management that can significantly reduce costs and improve operational efficiency.

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[Karabag/publication/378675717_Inventory_Management_Optimization_for_Intermittent_Demand/links/66033306a6d9fc55fd99215c/Inventory-Management-Optimization-forIntermittent-Demand.pdf](https://www.researchgate.net/profile/Oktay-Karabag/publication/378675717_Inventory_Management_Optimization_for_Intermittent_Demand/links/66033306a6d9fc55fd99215c/Inventory-Management-Optimization-forIntermittent-Demand.pdf)

The findings from this study are particularly pertinent to our initiative, which aims to create a web and mobile-based ordering, sales, and inventory management system for Mr. Siklo Motor Parts and Gears Trading. The focus on effective inventory strategies and precise demand forecasting aligns with our objective of streamlining stock level monitoring and establishing reorder notifications. The issues highlighted regarding the management

of intermittent demand for consumer goods parallel our focus on motor parts, where demand can fluctuate unpredictably. Furthermore, the incorporation of analytical techniques and the potential for leveraging big data analytics in the study indicate that our system could utilize gathered data for enhanced inventory management practices. By implementing similar approaches, we aspire to improve customer service and refine inventory management operations, ultimately leading to greater operational efficiency and customer satisfaction.

Kaya, B., Karabağ, O., Çekiç, F. R., Torun, B. C., Başay, A. Ö., Işıklı, Z. E., & Çakır, Ç. (2023, October). Inventory Management Optimization for Intermittent Demand. In *The International Symposium for Production Research* (pp. 768-782). Cham: Springer Nature Switzerland.

Advances in Spare Parts Classification and Forecasting for Inventory Control: A Literature Review

The study of Swapnil Bhalla et al. (2021) examines recent developments in spare parts management research, focusing on the integration of classification, forecasting, and inventory control. The review assesses the extent to which recent research addresses knowledge gaps previously identified in the field. The study highlights the importance of spare parts management for Original Equipment Manufacturers (OEMs) in supporting after-sales services and minimizing equipment downtime. The research identifies that most gaps highlighted in previous studies persist, indicating a continued need for integrated approaches to spare parts management. The review finds that the majority of recent spare parts management research takes the perspective of asset or equipment owners, with fewer case studies focusing on OEMs. The study also notes that most recent case studies are from the aviation, domestic appliance, and automotive industries. The authors emphasize the complexity of spare parts management systems, which require consideration of various factors such as demand volume, criticality, cost, and lead times.

Retrieved from:

<https://ntnuopen.ntnu.no/ntnuxmlui/bitstream/handle/11250/2829577/1-s2.0-S2405896321008703main.pdf?sequence=4>

The study implies to Mr. Siklo Motor Parts and Gears Trading, it provides valuable insights into the current state of spare parts management literature and highlights persistent research gaps. The emphasis on the need for integrated approaches to spare parts management, combining classification, forecasting, and inventory control, is particularly relevant to Mr. Siklo's work on developing an inventory management system. Understanding these trends and research gaps can guide Mr. Siklo in designing a more comprehensive and effective inventory management solution that addresses the unique challenges of spare parts, potentially incorporating advanced classification and forecasting techniques. This knowledge can lead to the development of a system that better balances service levels and costs, ultimately improving the company's operational efficiency and customer satisfaction in the context of spare parts management.

Bhalla, S., Alfnes, E., Hvolby, H. H., & Sgarbossa, F. (2021). Advances in spare parts classification and forecasting for inventory control: A literature review. *IFAC PapersOnLine*, 54(1), 982-987.

Sales Management System in Grocery Store Mobile Phone Based

According to Ardiyansyah and Fitrani (2021), the E-KELONTONG application was developed to enhance the sales management processes in traditional grocery stores, specifically targeting the challenges faced by An Nur Shop in Sidoarjo, Indonesia. The research outlines the operational difficulties due to frequent stock shortages stemming from high customer demand, necessitating a more efficient system for ordering goods. The application, created using Android Studio 3.5 and XAMPP server, aims to improve communication between the grocery store and sales representatives, thereby streamlining the ordering process and ultimately assisting in better inventory management. The study

emphasizes the importance of direct data collection through interviews and observations, ensuring that the application effectively addresses the specific needs of the grocery store.

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<https://download.garuda.kemdikbud.go.id/article.php?article=2916271&val=25639&title=Sales%20Management%20System%20in%20Grocery%20Store%20Mobile%20Phone%20Based>

In relation to our study on Mr. Siklo Motor Parts and Gears Trading, the findings of Ardiyansyah and Fitrani are particularly pertinent as they highlight the significance of effective inventory management and communication in retail operations. Both projects focus on implementing mobile technology to facilitate order processing and inventory tracking, albeit in different sectors. The E-KELONTONG application's approach to integrating sales management with a mobile platform resonates with our goal of developing a web and mobile-based system for Mr. Siklo that automates inventory management and order placement. Furthermore, the emphasis on improving supplier communication in the E-KELONTONG study aligns with our aim to enhance customer service and operational efficiency at Mr. Siklo, suggesting that the principles derived from the grocery sector can be effectively adapted to the motor parts industry for similar benefits.

Ardiyansyah, M. F., & Fitrani, A. S. (2021). Sales Management System in Grocery Store Mobile Phone Based Sistem Manajemen Penjualan di Toko Kelontong Berbasis Ponsel. *Procedia of Engineering and Life Science Vol, 1*(1).

Inventory management system

According to Pawar et al. (2024), inventory management systems are crucial for the success and sustainability of general stores. The researchers emphasize the importance of effective inventory management in maintaining optimal stock levels, reducing stockouts and overstocking, and enhancing overall operational efficiency. The study highlights

several key benefits of implementing inventory management systems, including real-time visibility into inventory levels, cost control through optimized inventory investment, improved operational efficiency through task automation, enhanced customer satisfaction, data-driven decision-making capabilities, and assistance in regulatory compliance. Pawar et al. (2024) proposes a comprehensive software solution to address common challenges faced by general store owners, such as inaccurate tracking, lack of real-time visibility, inefficient stock replenishment, and data management complexity.

Retrieved from: <https://doi.org/10.48175/IJARSCT-17288>

The findings of Pawar et al. (2024) are highly relevant to our study on Mr. Siklo Motor Parts and Gears Trading. While their research focuses on general stores, the core principles and benefits of inventory management systems are applicable to Mr. Siklo's business in the motor parts industry. The emphasis on real-time inventory tracking, automated stock replenishment, and data-driven decision-making aligns closely with our project's objectives for Mr. Siklo's web and mobile-based ordering, sales, and inventory management system. Furthermore, the challenges identified by Pawar et al. (2024), such as inaccurate tracking and lack of real-time visibility, mirror those faced by Mr. Siklo, underscoring the potential value of implementing a robust inventory management solution. The study's focus on user-friendly interfaces, customizable reporting, and system integration also resonates with our project's goals, providing valuable insights for the development and implementation of Mr. Siklo's new system.

Pawar, M. J., Shinde, M. S., Salunkhe, A. S., & Yadav, R. M. (2024). Inventory management system. International Journal of Advanced Research in Science Communication and Technology.

Local Study

Analysis and design of a sales and inventory management information system for a motorcycle parts and accessories store

The study by Magallanes et al. (2021) explores the development of a comprehensive inventory management system tailored for LJJG Motorcycle Parts and Accessories. The research emphasizes the importance of efficient inventory control, highlighting that traditional manual methods often lead to inaccuracies and inefficiencies in managing stock levels. By implementing an automated system, the study aims to enhance inventory tracking, reduce the risk of stockouts, and streamline the sales process. The proposed system features functionalities such as real-time inventory monitoring, automated restocking notifications, and detailed sales reporting, which collectively contribute to improved operational efficiency and customer service.

Retrieved from:

https://www.researchgate.net/publication/352978619_ANALYSIS_AND_DESIGN_OF_A_SALES_AND_INVENTORY_MANAGEMENT_INFORMATION_SYSTEM_FOR_A_MOTORCYCLE_PARTS_AND_ACCESSORIES_STORE

In relation to Mr. Siklo Motor Parts and Gears Trading, this study underscores the necessity for small businesses to transition from computerized inventory management practices to automated systems. Similar to LJJG, Mr. Siklo's reliance on outdated methods can lead to significant challenges, including inventory inaccuracies and employee theft. By adopting a modern inventory management system as proposed in this research, Mr. Siklo can enhance inventory oversight, reduce operational costs, and ultimately improve customer satisfaction. The insights from Magallanes et al. provide a valuable framework for developing a tailored solution that meets the specific needs of Mr. Siklo's business, thereby strengthening its competitive position in the market.

Magallanes, C. A., Ortiz, M. N., Seville, M. N., Luke, S., Tejada, G., Tuliao, E. M.,

Buladaco, M. V. M., & Eroy, N. G. (2021). Analysis and design of a sales and inventory management information system for a motorcycle parts and accessories store. *International Journal of Scientific Research and Engineering Development*, 4(3), 1407-1417.

Point-of-Sale Solution with Built-in CRM and Appointment Scheduling Empowered by Android Mobile Application

In regard to the study titled "Point of Sale Solution with Built-in CRM and Appointment Scheduling Empowered by Android Mobile Application" by Maurat et al. (2024) introduces a comprehensive web and mobile-based POS system designed to enhance business operations through features like appointment scheduling, CRM, and real-time inventory management. The system streamlines appointment management by allowing customers to schedule services easily via a mobile app or website, reducing manual coordination and improving operational efficiency. It also supports various payment methods, including cash and digital wallets such as GCash and Maya, providing flexibility and convenience to customers. Additionally, the system's real-time inventory management helps businesses monitor stock levels, receive low-stock alerts, and prevent losses, thus improving inventory control. The built-in CRM feature allows businesses to strengthen customer relationships by tracking purchase histories and analyzing customer behavior, enabling more personalized marketing strategies that enhance customer satisfaction and loyalty.

Retrieved from: <https://ieeexplore.ieee.org/abstract/document/10668385/>

The study implies to Mr. Siklo Motor Parts and Gears Trading, the study implies that integrating a POS system with features like real-time inventory tracking and customer relationship management can significantly benefit the business. Similar to the issues faced by other small businesses relying on computerized processes, Mr. Siklo can streamline its operations by adopting a modern, technology-driven solution. The use of digital payment options, automated inventory management, and customer tracking features can enhance the business's operational performance, reduce human errors, and boost customer satisfaction, aligning well with the goals of developing an efficient inventory and sales system for Mr. Siklo.

Maurat, J. I. C., Isip, E. V., Santillan, J. A., Lumabas, A. G. R., Ramirez, E. A., & Primo,

R. C. (2024, August). Point-of-Sale Solution with Built-in CRM and Appointment

Scheduling Empowered by Android Mobile Application. In 2024 IEEE 6th Symposium on Computers & Informatics (ISCI) (pp. 91-94). IEEE.

Angels and Lemons' E-Ordering: Development of a Web-Based Ordering System for Angels and Lemons.

Catubag et al. (2024) discuss the creation of "Angels and Lemons' E-Ordering," a web-based ordering platform tailored for Angels and Lemons, a lemonade business. This system is designed to improve the ordering process, minimize order-taking errors, and enhance the overall customer experience. Notable features include a carting mechanism for orders, prototypes of payment gateways like G-Cash and PayMaya to reduce the risk of fraudulent buyers, and an interface that connects customers, staff, and administrators for effective order fulfillment. Customers can place orders through the website or in-person, with options to select their preferred pickup times for online orders. The staff is tasked with preparing and fulfilling all orders, accommodating both upfront cash payments and e-wallet transactions. Administrative functions include managing products (addition, editing, and archiving). The system also produces vital business reports, including sales data, product performance trends, and comparisons of order methods. The study highlights the benefits of implementing a web-based ordering system to improve order accuracy and streamline operations for small businesses. It is worth noting that this system was developed for demonstration purposes and is not meant for actual operational use.

Retrieved from:

https://www.researchgate.net/profile/Rozel_Avila/publication/381881813_Angels_and_Lemons'_E-Ordering_Development_of_a_Web-Based_Ordering_System_for_Angels_and_Lemons/links/668367740a25e27fbc1d31a1/Angels-and-Lemons-E-Ordering-Development-of-a-Web-Based-Ord

The study implies to Mr. Siklo Motor Parts and Gears Trading, the study implies that adopting a similar web-based system could significantly improve operational

efficiency. Just as Angels and Lemons benefited from reduced order errors and enhanced customer experiences, Mr. Siklo could implement an automated system to optimize its inventory and order management, reduce human error, and better serve its customers. Integrating such technology could empower Mr. Siklo to make data-driven decisions and adapt to the evolving demands of the market.

Catubag, J. L. C., Fabian Jr, J. D., Garcia, E. J., Avila, R. B., & Cabance, P. J. Angels and Lemons' E-Ordering: Development of a Web-Based Ordering System for Angels and Lemons.

Dynamic Online Ordering and Data-Driven Inventory Management System with SMS for Security

The study by Bermusa et al. (2020) introduces a web-based ordering system aimed at addressing the challenges associated with manual inventory processes. In their research, the system allows customers to create accounts, browse products, add items to a shopping cart, and submit orders online, while administrators manage inventory, approve or cancel orders, and send SMS notifications to customers. The study highlights the system's capability to improve efficiency through real-time inventory updates, order tracking, and automated notifications, ensuring that business operations run smoothly and securely.

Retrieved from: [https://www.researchgate.net/profile/Edward-](https://www.researchgate.net/profile/Edward-Panganiban/publication/363796915_Dynamic_Online_Ordering_and_Data-Driven_Inventory_Management_System_with_SMS_for_Security/links/632e4acc86b22d3db4d9d467/Dynamic-Online-Ordering-and-Data-Driven-Inventory-ManagementSystem-with-SMS-for-Security.pdf)

[Panganiban/publication/363796915_Dynamic_Online_Ordering_and_Data-](https://www.researchgate.net/profile/Edward-Panganiban/publication/363796915_Dynamic_Online_Ordering_and_Data-Driven_Inventory_Management_System_with_SMS_for_Security/links/632e4acc86b22d3db4d9d467/Dynamic-Online-Ordering-and-Data-Driven-Inventory-ManagementSystem-with-SMS-for-Security.pdf)

[Driven_Inventory_Management_System_with_SMS_for_Security/links/632e4acc86b22d3db4d9d467/Dynamic-Online-Ordering-and-Data-Driven-Inventory-ManagementSystem-with-SMS-for-Security.pdf](https://www.researchgate.net/profile/Edward-Panganiban/publication/363796915_Dynamic_Online_Ordering_and_Data-Driven_Inventory_Management_System_with_SMS_for_Security/links/632e4acc86b22d3db4d9d467/Dynamic-Online-Ordering-and-Data-Driven-Inventory-ManagementSystem-with-SMS-for-Security.pdf)

In relation to Mr. Siklo Motor Parts and Gears Trading, the study provides critical insights into how transitioning from manually inputting data in Excel to automated inventory and sales systems can streamline business operations. Similar to the issues identified in Bermusa et al. study, Mr. Siklo Motor Parts and Gears Trading faces

challenges related to sales tracking and inventory management, which lead to errors and inefficiencies. The integration of a web and mobile-based system, as demonstrated in the dynamic online ordering and inventory system, can significantly enhance Mr. Siklo's ability to manage inventory, prevent stockouts, and improve customer satisfaction through email notifications. This underscores the importance of adopting technological solutions to meet the needs of modern businesses, particularly those seeking to optimize their sales and inventory processes for better operational performance.

Bermusa, J. P., Bermusa, M. C. M., Macayanan, K. C., Panganiban, E. B., Abad Jr, B.

B., Bogalin, V. S., ... & Claravall, J. L. O. (2020). Dynamic Online Ordering and Data-Driven Inventory Management System with SMS for Security. *International Journal*, 8(9).

Leyte Normal University: Supply Inventory Management System

The study of Anade et al. (2023) investigates the current inventory management practices at Leyte Normal University in the Philippines. The research begins with a thorough assessment of the existing system, focusing on procurement, stock control, and record-keeping processes to pinpoint inefficiencies. Through interviews, surveys, and data analysis, the authors identify critical challenges such as stockouts, overstocking, inaccurate records, and delays in procurement. Based on these findings, the study proposes enhancements to the inventory management system, which may include technological integration, process optimization, staff training, and standardized inventory control procedures. Furthermore, the researchers evaluate the potential impact of these enhancements using performance metrics like inventory turnover rate, stock accuracy, and procurement lead time, aiming to quantify the benefits and return on investment.

Retrieved from:

https://www.researchgate.net/publication/374753747_Leyte_Normal_University_Supply_Inventory_Management_System

In relation to Mr. Siklo Motor Parts and Gears Trading, this study highlights the importance of a systematic approach to inventory management. Similar to the challenges faced by Leyte Normal University, Mr. Siklo may encounter issues related to inaccurate stock records and inefficient procurement processes due to reliance on manually inputting data through Excel. By implementing a more advanced inventory management system as suggested in this research, Mr. Siklo can streamline operations, reduce costs, and ensure better availability of motorcycle parts and accessories, ultimately enhancing customer satisfaction and competitive advantage in the market.

Anade, M. A. M., Calibo, C. T. C., Noveda, J. P., Wendell, R. Y., & Verecio, R. L.

(2023). Leyte Normal University: Supply inventory management system.

International Journal of Innovative Science and Research Technology, 8(10).

Foreign Studies

Mobile Web-Based Cassava Inventory Management System: A Case Study at CV Panca Yoga

According to the study of Haris Andika Pratama and Sri Wulandari (2023), the inefficiencies in inventory management at CV Panca Yoga, a company engaged in the cassava trading industry, are highlighted. The research emphasizes the transition from traditional manual bookkeeping methods to a digitized web and mobile-based system designed to streamline inventory processes, including the management of incoming and outgoing goods, stock inventory, and report generation. Utilizing a quantitative research methodology, the authors collected data through direct observation and structured questionnaires, which revealed significant challenges in the existing computerized processes, including human error and inefficiencies. The newly developed system not only enhances operational efficiency but also reduces the potential for inaccuracies in

inventory management, demonstrating the critical role of technology in modern business practices.

Retrieved from:

<https://www.journal.lembagakita.org/index.php/ijsecs/article/view/1866/1346>

In relation to Mr. Siklo Motor Parts and Gears Trading, the study provides valuable insights into the challenges faced by businesses that rely on computerized methods for tracking inventory. Many businesses, particularly those like Mr. Siklo's that still use traditional tools such as Microsoft Excel, struggle with inefficiencies and inaccuracies in their inventory processes. This highlights the importance of integrating automated systems to enhance operational effectiveness and data accessibility. Understanding these challenges can inform the development of a modern inventory management system tailored to meet the specific needs of businesses transitioning from computerized to automated solutions. By adopting technology-driven approaches, Mr. Siklo can improve operational performance, increase profitability, and enhance competitiveness in today's fast-paced market, ultimately leading to greater efficiency and customer satisfaction.

Pratama, H. A., & Wulandari, S. (2023). Mobile Web-Based Cassava Inventory

Management System: A Case Study at CV Panca Yoga, Purbalingga. *International Journal Software Engineering and Computer Science (IJSECS)*, 3(3), 467-473.

Online Restaurant Management and Food Ordering System (ORMFOS)

The study of Weerasekara (2021) presents a web-based application designed to enhance restaurant management through various services, including food ordering, table reservations, customer information management, waiter information management, menu management, and reporting. The research identifies significant limitations in the existing system, which is characterized by inefficiencies, excessive time consumption, and high operational costs. Key issues include data entry inconsistencies, costly staff training, inadequate information sharing, and overall ineffectiveness. The proposed system aims to

streamline daily operations, making them more time-efficient and reducing errors. Specific objectives include minimizing employee numbers, lowering labor costs, preventing long queues, enabling custom meal orders, providing ingredient information, improving client-server communication, and automating bill computations and report generation. The development process utilized a Rapid Application Development (RAD) approach. Feedback from domain and software development experts indicated positive evaluations regarding the system's appearance, usability, functionality, and performance.

Retrieved from: <https://dl.ucsc.cmb.ac.lk/jspui/handle/123456789/4289>

The study implies to Mr. Siklo Motor Parts and Gears Trading, this study emphasizes the pressing need for businesses that depend on manual coordination processes of inventory management and sales tracking to automated systems. Similar to the challenges faced by the restaurant industry, Mr. Siklo's reliance on Microsoft Excel for inventory management leads to inefficiencies and potential inaccuracies. By implementing a web-based ordering and inventory management system, Mr. Siklo can enhance operational efficiency, minimize human error, and improve customer satisfaction. The insights from the ORMFOS study can inform the development of a tailored solution that addresses the specific needs of Mr. Siklo's business, ultimately fostering better inventory management practices and a more competitive position in the market.

Weerasekara, U. S. (2021). Online Restaurant Management and Food Ordering System for ORMFOS (Doctoral dissertation).

Inventory Management Systems (IMS)

According to Chin, Ramiah, and Razali (2023), this study aims to develop an Inventory Management System (IMS) that enhances control over product stock, customer orders, service, and order delivery for small and medium enterprises (SMEs) in Malaysia. The research emphasizes that many SMEs still rely on manual inventory management, leading to inefficiencies. By utilizing PHP for backend development and HTML, CSS, and

JavaScript for the frontend, the study adopts a Rapid Application Development (RAD) methodology, which highlights the iterative process of development. Although the IMS has been successfully developed, the authors note existing limitations and suggest areas for future enhancements.

Retrieved from:

https://www.researchgate.net/publication/372653862_Inventory_Management_Systems_IMS

In relation to Mr. Siklo Motor Parts and Gears Trading, the findings of this study are particularly relevant. The tracking methods currently employed may lead to similar inefficiencies and inaccuracies as experienced by the SMEs in the study. Implementing a more automated IMS could provide Mr. Siklo with improved inventory tracking, better customer order management, and overall operational efficiency, thereby enhancing customer satisfaction and business performance.

Chin, C., Ramiah, S., & Razali, N. F. (2023). Inventory Management Systems (IMS).

Journal of Technology and Management. 2600-7304.

Point of Sale and Cashless Policy in Nigeria: Challenges and Prospects

The study of Anyanwu and Anumaka (2020) investigates the influence of Point of Sale (POS) systems on the implementation of the cashless policy in Nigeria. The research highlights that the consistent use of POS terminals has a significant and positive impact on the adoption of cashless transactions. However, it also identifies that syntax errors in POS transactions, such as over-debiting and reversal errors, can negatively affect the acceptance of this policy. Furthermore, the study notes that while POS usage is important, other factors—such as infrastructure, public awareness, and the regulatory environment—play a more crucial role in the overall success of the cashless policy. Employing econometric techniques like co-integrating least squares, autoregressive

distributed lag, and Granger causality, the authors provide empirical evidence of the relationship between POS systems and the cashless initiative. The findings offer valuable insights for policymakers and financial institutions aiming to enhance the effectiveness of the cashless policy in Nigeria.

Retrieved from: <http://info.openarchivelibrary.com/id/eprint/329/>

In relation to Mr. Siklo Motor Parts and Gears Trading, this study underscores the importance of integrating reliable payment systems to facilitate smoother transactions. Just as the POS systems are essential for the cashless policy, Mr. Siklo can benefit from adopting modern payment solutions to improve operational efficiency and customer satisfaction. By addressing potential transaction errors and enhancing payment processing, Mr. Siklo can streamline operations, reduce human error, and foster a more positive customer experience, ultimately leading to a stronger competitive position in the market.

Anyanwu, G. I., & Anumaka, C. I. (2020). Point of Sale and Cashless Policy in Nigeria: Challenges and Prospects. *Asian Journal of Economics, Business and Accounting*, 20(2), 42-57.

Implementing an Automated Inventory Management System for Small and Medium-sized Enterprises

The study of Salih, Ghazi, and Aljanabi (2023) investigates the deployment of an automated inventory management system at Alfa Company, a small to medium-sized enterprise in Iraq. The research highlights several key benefits of the system, including enhanced inventory tracking that allows for real-time monitoring of stock levels, which reduces the risk of stock outs and excess inventory. Furthermore, the implementation of the AI-driven system led to reduced capital investment and holding costs by optimizing inventory levels, ultimately improving the financial performance of the company. The data analytics and demand forecasting capabilities of the system provided valuable insights for

effective inventory management, enabling informed decisions regarding stock replenishment. However, the study also acknowledges challenges during the implementation phase, such as the technological complexity of the system and the necessity for employee training. Overall, the automated inventory management system has streamlined Alfa Company's inventory procedures, reduced costs, and improved customer service, positioning the company as a market leader in the distribution of laboratory equipment and supplies in Iraq.

Retrieved from: <https://www.iasj.net/iasj/download/744695d223188a72>

This study implies to Mr. Siklo Motor Parts and Gears Trading, this study underscores the transformative potential of automated inventory management systems for small to medium-sized enterprises. Similar to Alfa Company, Mr. Siklo could greatly benefit from enhanced inventory tracking and data-driven decision-making to optimize stock levels and reduce holding costs. By transitioning from traditional inventory management methods to an automated system, Mr. Siklo can improve operational efficiency, minimize human error, and ultimately enhance customer satisfaction, thereby strengthening its competitive position in the market.

Salih, H. S., Ghazi, M., & Aljanabi, M. (2023). Implementing an Automated Inventory Management System for Small and Medium-sized Enterprises. *Iraqi Journal For Computer Science and Mathematics*, 4(2), 238-244.

Chapter 3

RESEARCH METHODOLOGY

This chapter outlines the methods and processes employed to develop the Web and Mobile-Based Point-of-Sales System with Email Notification for Mr. Siklo Motor Parts and Gears Trading. It details the research approach, tools, and techniques for efficient development process.

Locale of the Study

This study is conducted at Mr. Siklo Motor Parts and Gears Trading, a business specializing in selling motor parts, gears, and accessories. The company operates in a retail setting, serving both walk-in customers and online buyers. The physical store is located in Marcos Highway, Antipolo, Philippines, where the daily operations of sales and inventory management take place.

Currently, the business relies on Microsoft Excel for recording sales and manually tracking inventory. Due to the large number of products offered, challenges such as inaccurate stock monitoring and inefficient sales processing arise. This study aims to develop a web and mobile-based point-of-sales (POS) system with email notification to address these issues.

The study focuses on implementing the system within Mr. Siklo's business environment, testing it with store employees, management, and selected customers. The system will be evaluated based on its ability to automate inventory tracking, sales transactions, and order notifications, ultimately improving operational efficiency and customer satisfaction.

Respondents of the Study

The respondents of this study consist of individuals directly involved in the daily operations of Mr. Siklo Motor Parts and Gears Trading, as well as customers who engage in purchasing motor parts and accessories. These respondents were selected based on their role and interaction with the current sales and inventory management process. The following groups are identified as the key respondents:

1. Business Owner

- The owner is responsible for overseeing business operations, managing inventory, and processing sales transactions. Their insights are crucial in evaluating the effectiveness of the proposed system in improving operational efficiency.

2. Employees (Cashiers and Inventory Staff)

- Employees who handle sales transactions, inventory updates, and order processing will participate in the study. Their feedback will help assess how the system enhances workflow efficiency, minimizes errors, and simplifies inventory tracking.

3. Customers

- Selected customers who purchase motor parts and accessories from Mr. Siklo will be surveyed or interviewed to evaluate their experience with the new system, particularly in terms of ease of ordering, transaction processing, and email notifications.

Sampling Method

A purposive sampling method will be used to select respondents who have direct interaction with the business processes. This method ensures that feedback is gathered from individuals most affected by the current system and the proposed improvements.

Sample Size

The study will involve one business owner or manager, two to three employees handling sales and inventory, and 10 to 20 customers who have placed orders or made purchases. Their responses will give helpful insights into how the proposed web and mobile-based POS system with email notifications can address business challenges.

Methodology of the Study

The research methodology provides a structured approach to developing the proposed system. Following the principles of Agile methodology, this study employs iterative cycles to ensure continuous improvement, adaptability to requirements, and the delivery of a functional system that meets the business's needs.

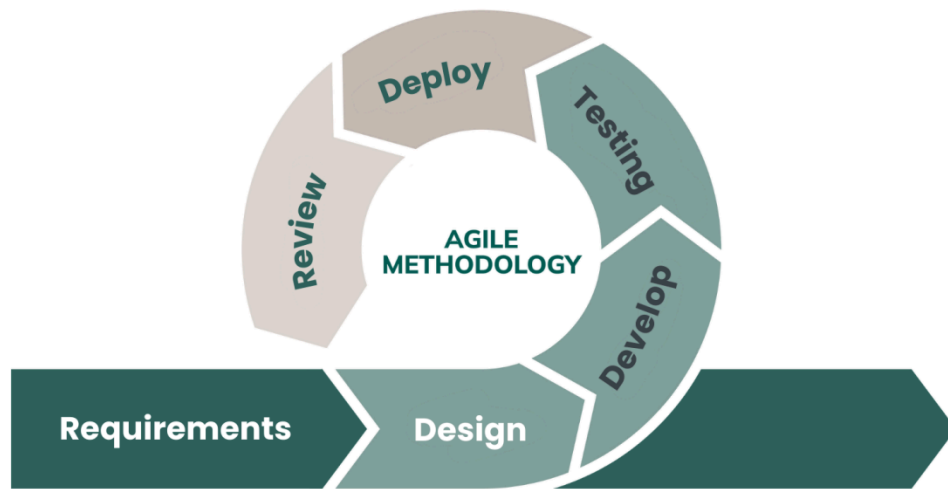


Figure 2. Agile Methodology

Figure 2 illustrates the Agile methodology model of Software Development Life Cycle (SDLC) adopted in this study. The process is divided into six key phases:

Phase 1 - Requirements

This phase focuses on identifying the scope of the problem and the necessary solution for the study. Data and system requirements are gathered through discussions with the business owner and staff of Mr. Siklo Motor Parts and Gears Trading. The goal is to evaluate and analyze primary needs, such as sales tracking, inventory management, customer engagement, and reporting. Continuous communication ensures a comprehensive understanding of the system requirements. Functional and non-functional requirements are documented, and system analysis is conducted to align the solution with business expectations. This phase concludes with a well-defined problem statement and the foundation for system design.

Phase 2 - Design

The design phase focuses on creating the system blueprint based on the gathered requirements. This includes planning the system architecture, database structure, and user interface. The design ensures that the system is user-friendly, functional, and responsive on both web and mobile platforms. Mock-ups and wireframes are developed to visualize the flow of processes such as product browsing, order placement, inventory tracking, and notifications. This phase bridges the gap between requirements and implementation, ensuring clarity for the development team.

Phase 3 - Development

This phase involves building the system's modules based on design specifications. The development team adopts an incremental approach, working in sprints focused on functionalities like sales processing, inventory management, and email notifications. Regular feedback loops ensure the system aligns with user requirements. Tools, programming languages, and technologies are selected based on the system's needs, and iterative cycles continue until all functionalities are integrated.

Phase 4 – Testing

The testing phase involves rigorous evaluation of the system to identify and fix bugs or issues before deployment. Developers conduct functionality testing to verify that all features work as intended, such as sales tracking accuracy, real-time inventory updates, and email notifications. Usability testing ensures that both the admin and user interfaces are intuitive and easy to navigate. Performance testing is also conducted to validate the system's stability, responsiveness, and ability to handle multiple users or transactions simultaneously.

Phase 5 – Deployment

Deployment involves making the system fully operational for end-users. This phase includes installing the system in the business environment, migrating data from old systems, ensuring all components integrate correctly, and training users to operate the system efficiently. Feedback from real-world usage is gathered to address any remaining issues and ensure smooth operation.

Phase 6 – Review

The review phase involves gathering feedback from the business owner, staff, and customers regarding system performance and usability. The review identifies areas for improvement or enhancements, ensuring the system meets its intended goals. Insights from this phase, such as suggestions for feature refinements or interface adjustments, are used to optimize the system further. This phase ensures that the system continues to align with business operations and customer needs.

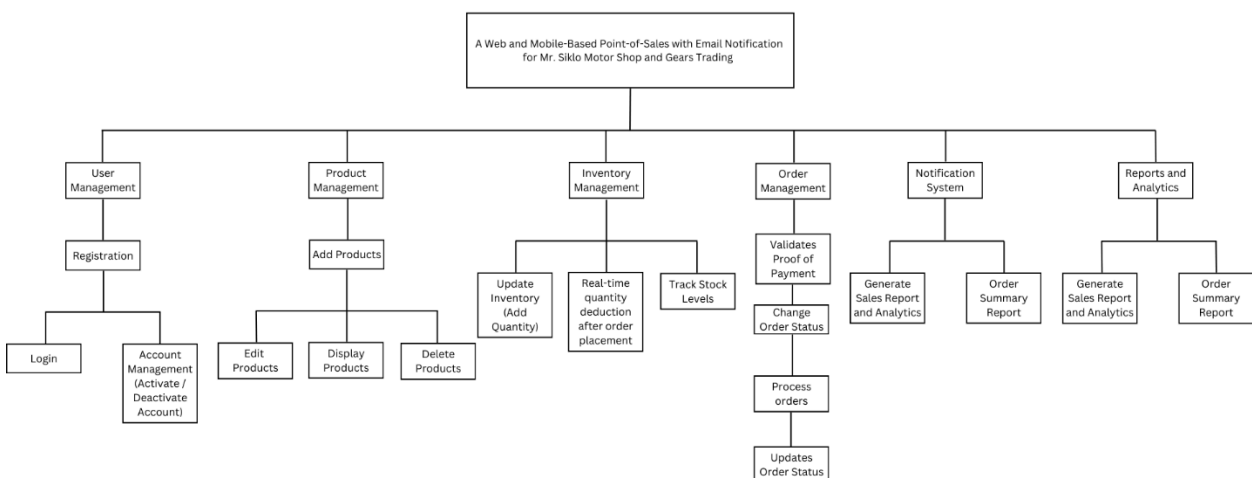


Figure 3. Hierarchical Input-Process-Output

The diagram is divided into major categories or modules of the system, with a clear hierarchy. Below is a breakdown of each main module and its corresponding subcomponents:

- **User Management:**

- Registration: Allows users to register for an account.
- Login: Enables users to access the system using their credentials.
- Account Management: Provides the ability to activate or deactivate user accounts.

- **Product Management:**

- Add Products: Facilitates the addition of new products to the inventory.
- Edit Products: Enables modification of product details.
- Display Products: Displays all available products.
- Delete Products: Allows removal of products from the inventory.

- **Inventory Management:**

- Update Inventory (Add Quantity): Lets users update stock levels by adding quantities.
- Real-Time Quantity Deduction After Order Placement: Automatically deducts stock when an order is placed.
- Track Stock Levels: Monitors and manages product stock availability.

- **Order Management:**

- Validates Proof of Payment: Ensures customer payments are verified.
- Change Order Status: Tracks and updates the status of orders.
- Process Orders: Manages order fulfillment.
- Updates Order Status: Keeps the system updated with the latest order information.

- **Notification System:**

- Generate Sales Report and Analytics: Produces detailed reports and analytics related to sales.
- Order Summary Report: Summarizes details of placed orders.

- **Reports and Analytics:**

- Generate Sales Report and Analytics: Provides in-depth analysis of sales trends.
- Order Summary Report: Displays summaries of orders for analysis and record-keeping.

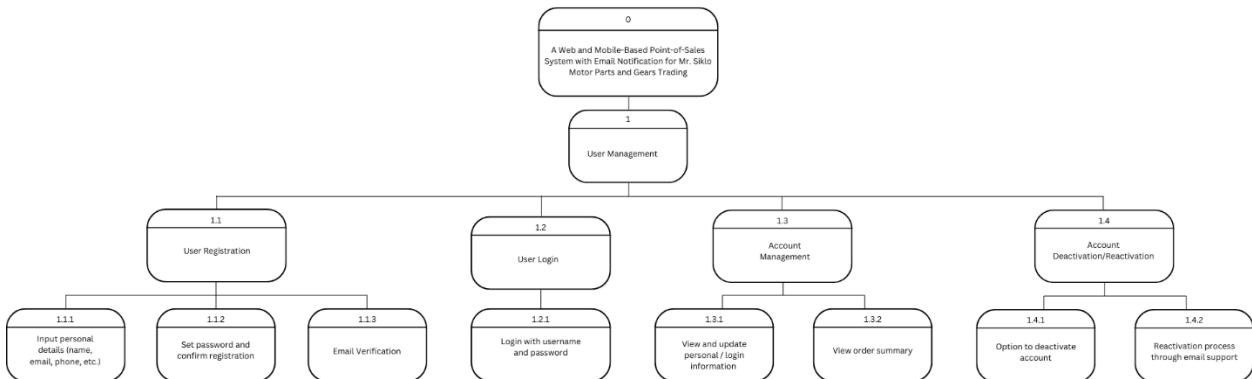


Figure 4. Virtual Table of Contents

Figure 4, presents a Virtual Table of Contents for a Web and Mobile-Based Point-of-Sales System with Email Notification for Mr. Siklo Motor Parts and Gears Trading. At the top level, the main system is identified, followed by the primary module, User Management, which handles all user-related functions. Under User Management, several sub-modules are detailed. User Registration includes processes such as inputting personal details (name, email, phone, etc.), setting a password and confirming registration, and email verification. The User Login sub-module covers logging in with a username and password. Account Management allows users to view and update personal or login information and view their order summary. Lastly, the Account Deactivation/Reactivation sub-module offers options to deactivate an account and provides a reactivation process through email support. This hierarchical structure offers an organized overview of the user-related components within the Point-of-Sales System.

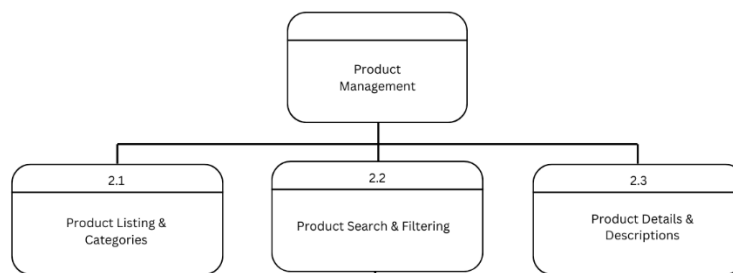


Figure 4.1 Virtual Table of Contents (Continuation)

Figure 4.1, the Virtual Table of Contents for the Web and Mobile-Based Point-of-Sales System with Email Notification for Mr. Siklo Motor Parts and Gears Trading, focusing on the Product Management module. Under this main module, three sub-modules are presented.

The first sub-module, Product Listing & Categories, includes a function to view available products such as helmets, gears, brackets, and intercoms. The second sub-module, Product Search & Filtering, allows users to search for products by name, category, and price range. Lastly, the Product Details & Descriptions sub-module provides access to product images, specifications, and pricing. This continuation further elaborates on the product-related functionalities within the Point-of-Sales System, complementing the user management section from the previous figure.

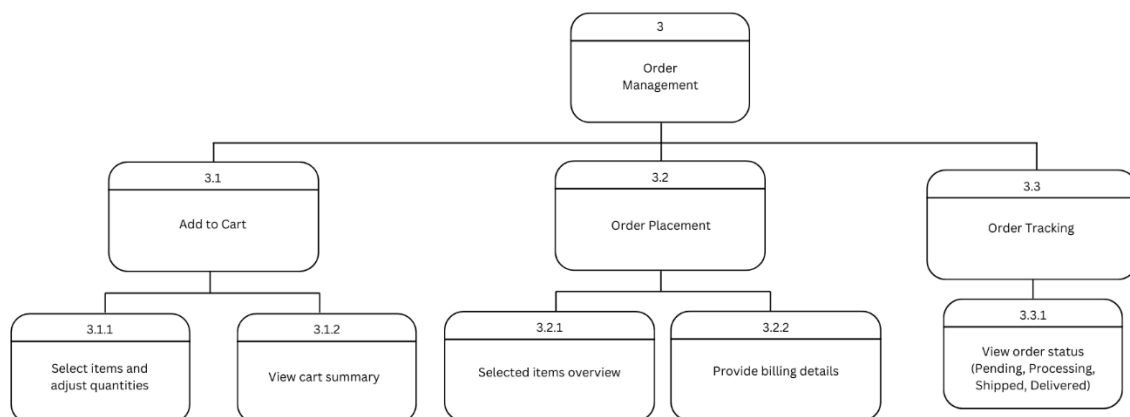


Figure 4.2 Virtual Table of Contents (Continuation)

Figure 4.2, continues the Virtual Table of Contents for the Web and Mobile-Based Point-of-Sales System with Email Notification for Mr. Siko Motor Parts and Gears Trading,

focusing on the Order Management module. The module is divided into three main sub-modules: Add to Cart, Order Placement, and Order Tracking.

The Add to Cart sub-module includes selecting items and adjusting quantities, along with viewing the cart summary. The Order Placement sub-module consists of an overview of selected items and providing billing details for order processing. Lastly, the Order Tracking sub-module allows users to view the status of their orders, categorized into pending, processing, shipped, and delivered. This continuation highlights the order-related functionalities within the Point-of-Sales System, completing the Virtual Table of Contents with user, product, and order management components.

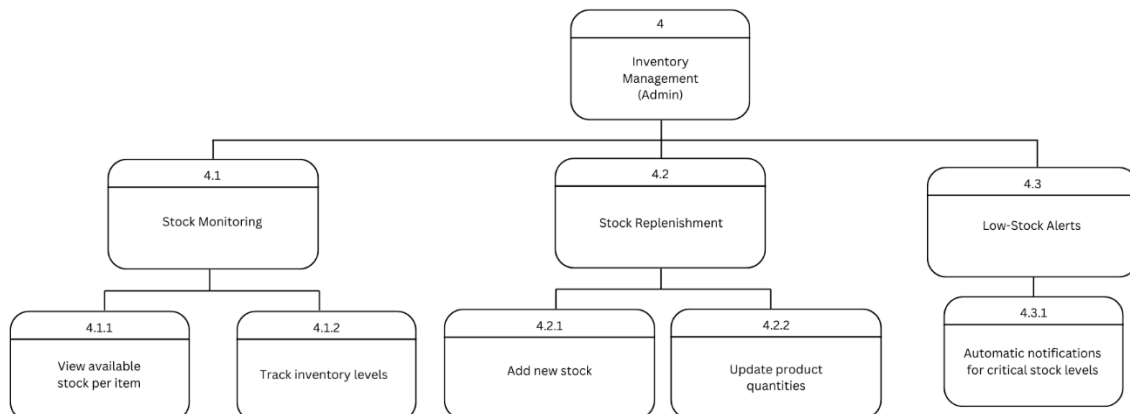


Figure 4.3 Virtual Table of Contents (Continuation)

Figure 4.3 presents a continuation of the Virtual Table of Contents, illustrating the hierarchical structure of the Inventory Management (Admin) module. This module is divided into three main sections: Stock Monitoring, Stock Replenishment, and Low-Stock Alerts. Stock Monitoring (4.1) focuses on tracking stock levels, including viewing available stock per item (4.1.1) and monitoring inventory levels (4.1.2). Stock Replenishment (4.2) involves managing stock updates, with functionalities for adding new stock (4.2.1) and updating product quantities (4.2.2). Lastly, Low-Stock Alerts (4.3) ensures inventory shortages are addressed by

sending automatic notifications for critical stock levels (4.3.1). This structured breakdown highlights the essential functions of inventory management, ensuring efficient stock tracking, replenishment, and timely alerts to maintain optimal inventory levels.

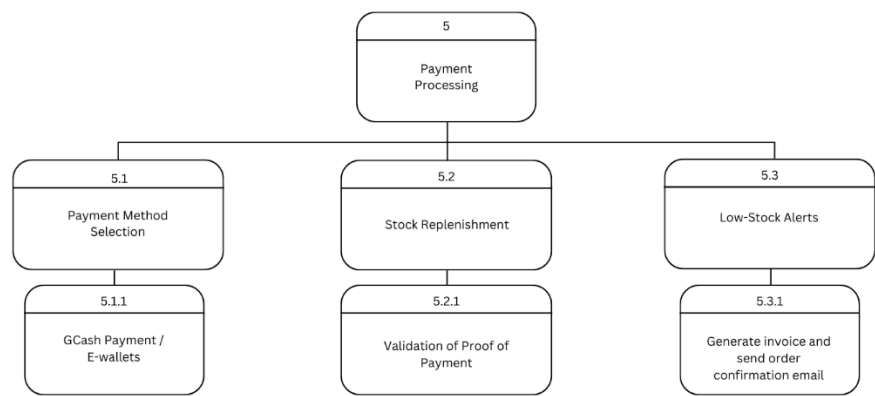


Figure 4.4 Virtual Table of Contents (Continuation)

Figure 4.4, presents a continuation of the Virtual Table of Contents, outlining the hierarchical structure of the Payment Processing module. This module is divided into three main sections: Payment Method Selection, Stock Replenishment, and Low-Stock Alerts. Payment Method Selection (5.1) focuses on choosing a preferred payment option, including GCash payments and e-wallet transactions (5.1.1). Stock Replenishment (5.2) ensures payments are verified before processing inventory updates, with validation of proof of payment (5.2.1) as a key function. Lastly, Low-Stock Alerts (5.3) includes generating invoices and sending order confirmation emails (5.3.1) to maintain transaction transparency. This structured breakdown ensures a smooth payment process by integrating various payment methods, verifying transactions, and automating essential notifications.

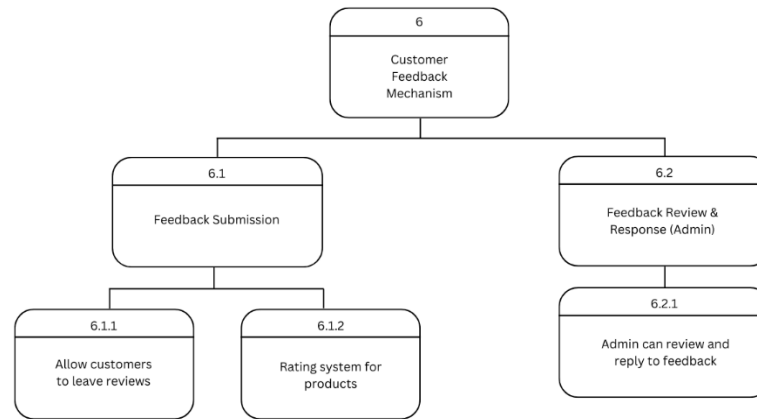


Figure 4.5 Virtual Table of Contents (Continuation)

Figure 4.5, presents a structured breakdown of the Customer Feedback Mechanism in a hierarchical format. At the top level, Section 6 represents the overall feedback system, which is divided into two main components: Feedback Submission (6.1) and Feedback Review & Response (Admin) (6.2). The Feedback Submission section focuses on gathering customer input and includes two subcategories: Allowing customers to leave reviews (6.1.1) and a Rating system for products (6.1.2). Meanwhile, the Feedback Review & Response (Admin) section highlights the role of administrators in managing feedback, with 6.2.1 specifying that admins can review and reply to customer feedback. This structured approach ensures efficient collection, evaluation, and response to customer feedback, ultimately improving service quality and user satisfaction.

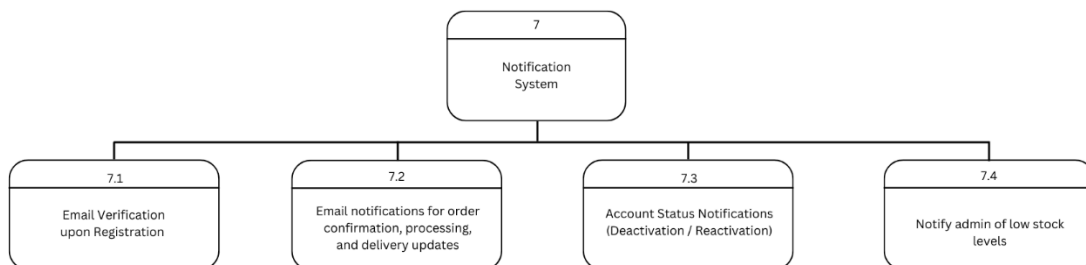


Figure 4.6 Virtual Table of Contents (Continuation)

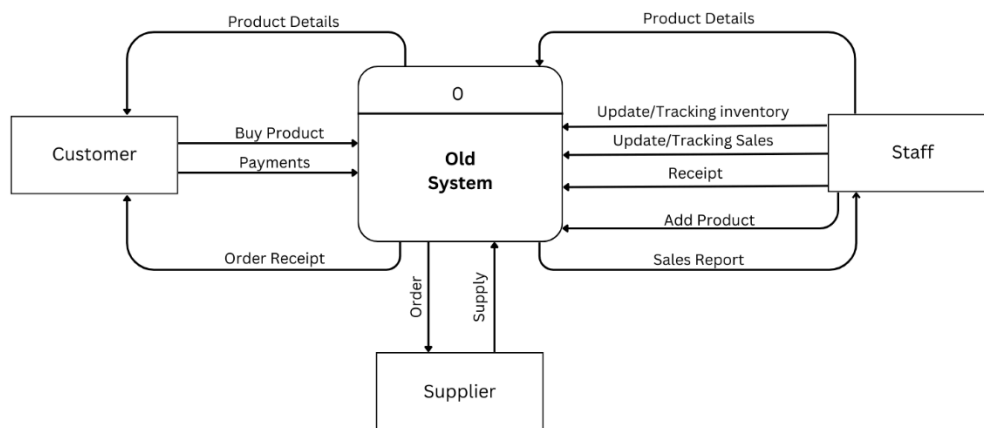
Figure 4.6, illustrates the structure of the Notification System in a hierarchical format. At the top level, Section 7 represents the Notification System, which is divided into

four key components. Email Verification upon Registration (7.1) ensures that users verify their email addresses during account creation. Email Notifications for Order Confirmation, Processing, and Delivery Updates (7.2) provide real-time updates to customers regarding their orders. Account Status Notifications (Deactivation/Reactivation) (7.3) inform users about changes in their account status. Lastly, Notify Admin of Low Stock Levels (7.4) helps administrators manage inventory efficiently by alerting them when stock levels are low. This notification system enhances communication, improves user experience, and streamlines administrative tasks.



Figure 4.7 Virtual Table of Contents (Continuation)

Figure 4.7, outlines the Reports & Analytics (Admin) module in a hierarchical structure. At the top level, Section 8 represents the Reports & Analytics functionality, which focuses on providing sales-related insights to administrators. This section is divided into Sales Reports (8.1), which further breaks down into two key components: Tracking daily, weekly, and monthly sales (8.1.1) to help administrators monitor revenue trends over different timeframes, and Viewing best-selling products (8.1.2) to identify high-demand items. This system enhances decision-making by providing valuable sales data, allowing administrators to optimize inventory and marketing strategies effectively.



Figure

5. Context Diagram (Old System)

Figure 5, illustrates the overall structure and interactions within the existing system. At the center, the Old System acts as the main processing unit, handling transactions and data exchanges between three key entities: Customers, Staff, and Suppliers.

Customers interact with the system by viewing Product Details, making Purchases, and processing Payments. In return, they receive an Order Receipt as confirmation. The Staff also engages with the system by managing Product Details, Updating/Tracking Inventory and Sales, generating Receipts, adding new products, and accessing Sales Reports. Meanwhile, the Supplier supplies products to the system upon receiving an Order, ensuring that stock levels are maintained.

This diagram highlights the manual and outdated processes of the old system, showcasing inefficiencies in tracking inventory, sales, and supply chain management. It serves as a basis for identifying areas of improvement in a proposed upgraded system.

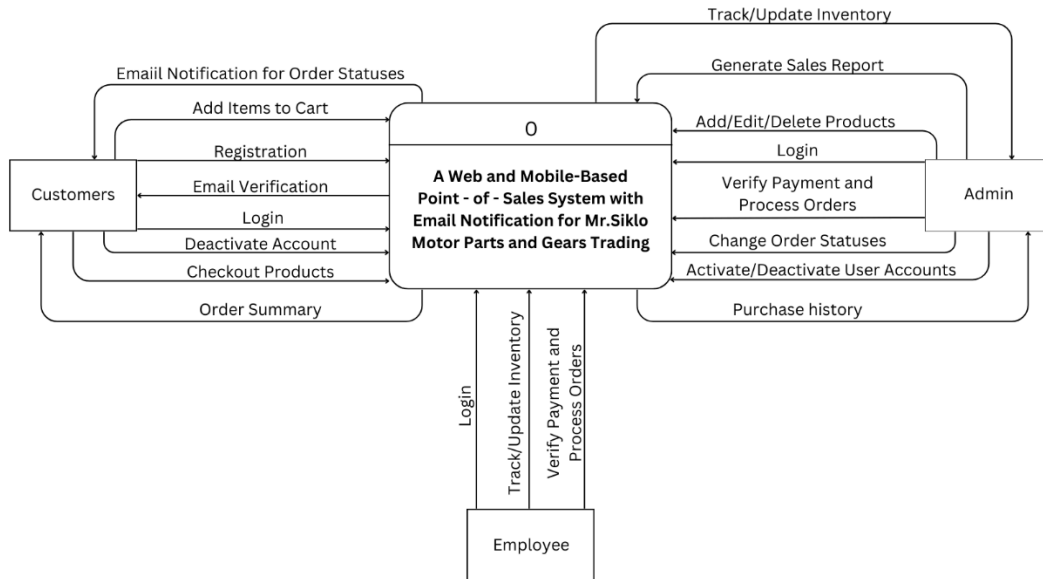


Figure 5.1 Context Diagram (Proposed System)

Figure 5.1, illustrates the improved Web and Mobile-Based POS System with Email Notifications for Mr. Siklo Motor Parts and Gears Trading. Customers can register, verify emails, log in, add items to cart, checkout, receive order updates, and manage accounts. Employees handle inventory tracking, payment verification, and order processing, while Admins manage inventory, sales reports, product updates, payments, orders, user accounts, and purchase history. This system enhances automation, reducing manual tasks and improving efficiency in sales, inventory, and customer interactions.

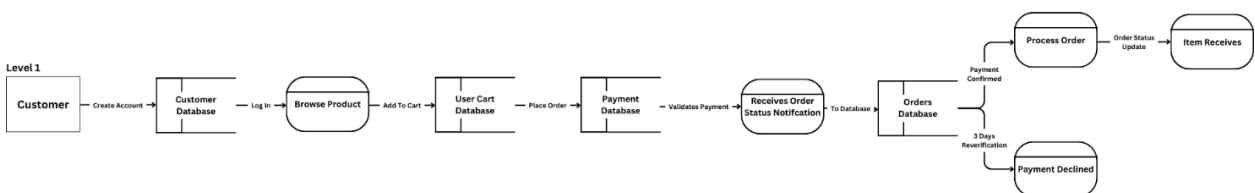


Figure 6. Data Flow Diagram (Level 1)

Figure 6, illustrates the structured process of a customer's interaction with the system. It begins with account creation and login, storing data in the Customer Database.

Customers can browse products, add items to their cart, and place orders, which are recorded in the User Cart Database. Payments are then processed through the Payment Database, validating transactions before updating the Orders Database. If payment is confirmed, the order is processed, and the customer receives an order status notification. Successful transactions lead to order fulfillment, while declined payments halt the process. A three-day re-verification ensures order accuracy, enhancing efficiency in order management, payment validation, and customer notifications.

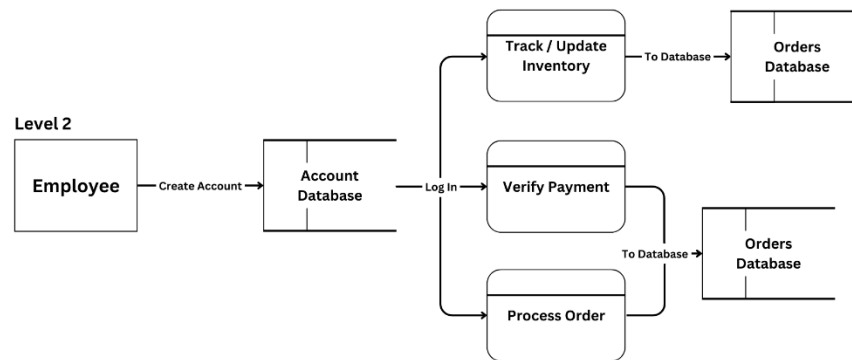


Figure 6.1 Data Flow Diagram (Level 2)

Figure 6.1, details the employee's interaction with the system. Employees begin by creating an account, which is stored in the Account Database. After logging in, they can perform key tasks such as tracking and updating inventory, verifying payments, and processing orders. These actions update the Orders Database, ensuring accurate record-keeping and efficient order management. This diagram highlights the workflow for employees in managing inventory, validating transactions, and overseeing order fulfillment.

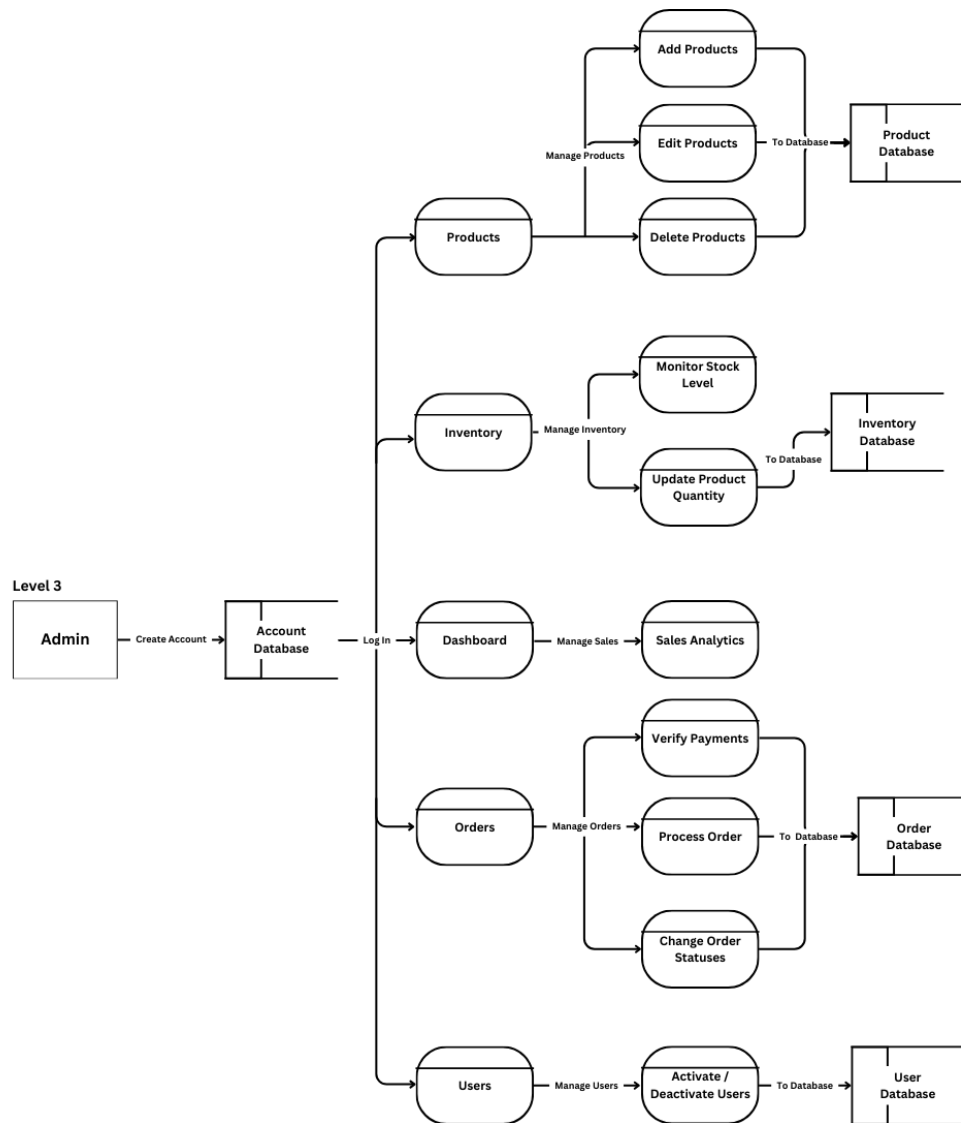


Figure 6.2 Data Flow Diagram (Level 3)

Figure 6.2, illustrates the admin's role in managing the system. After creating an account and logging in through the Account Database, the admin can handle products, inventory, orders, users, and sales analytics via the Dashboard. Admins can add, edit, or delete products, monitor stock levels, and update inventory. They also verify payments, process orders, and change order statuses, ensuring smooth transactions stored in the Order Database. Additionally, admins can activate or deactivate user accounts and analyze sales data for decision-making. This diagram highlights the comprehensive control admins have over system operations

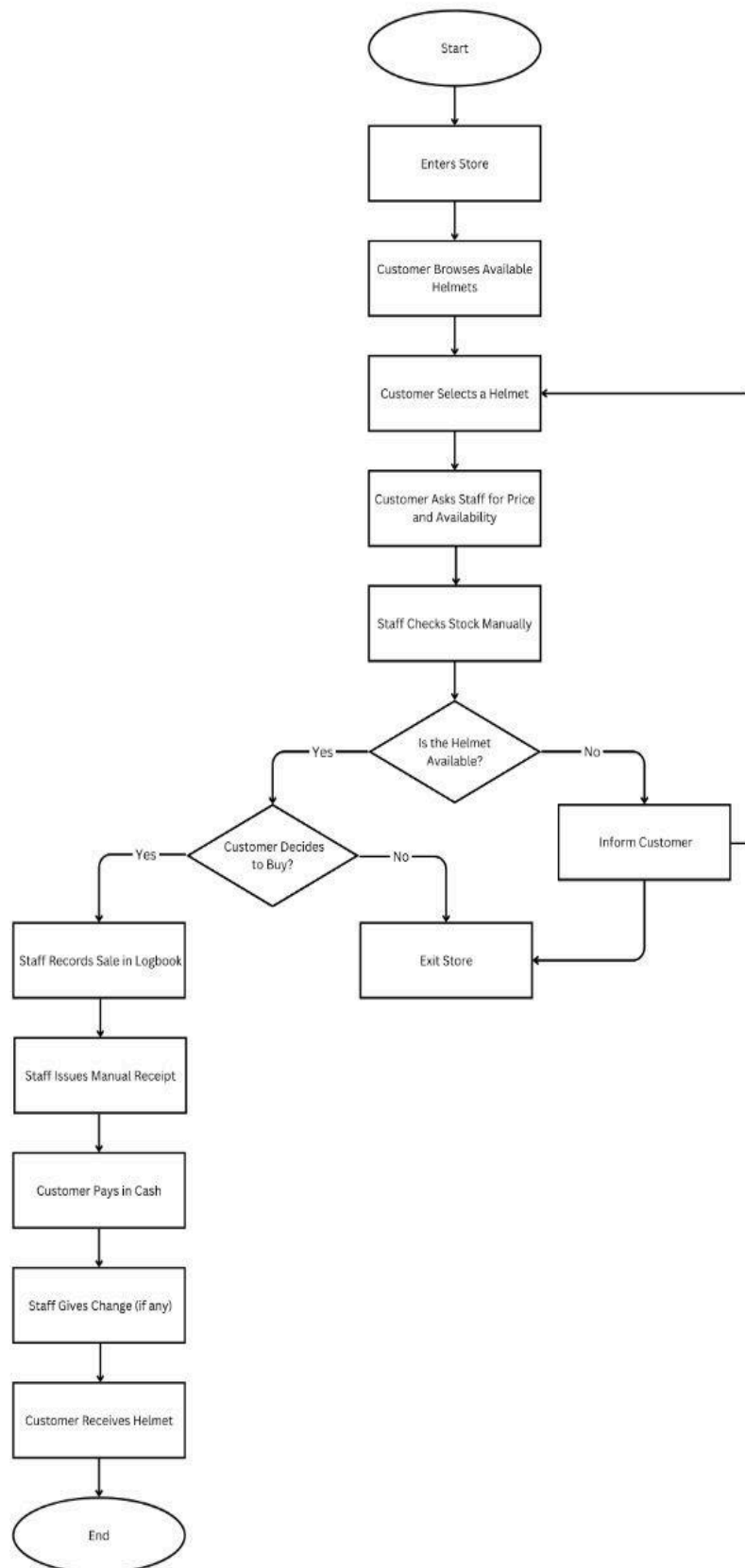


Figure 7. System Flowchart (Old System)

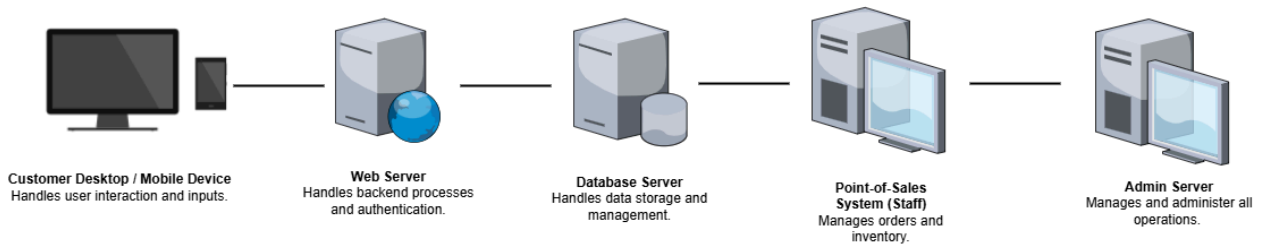


Figure 7. Architectural Design

Figure 7, presents the system's infrastructure, illustrating the interaction between components. Customers use desktop or mobile devices to interact with the system, which sends requests to the web server for backend processing and authentication. The database server handles data storage and management, ensuring secure access. The point-of-sales system allows staff to manage orders and inventory, while the admin server oversees and administers all operations, ensuring smooth functionality across the system.

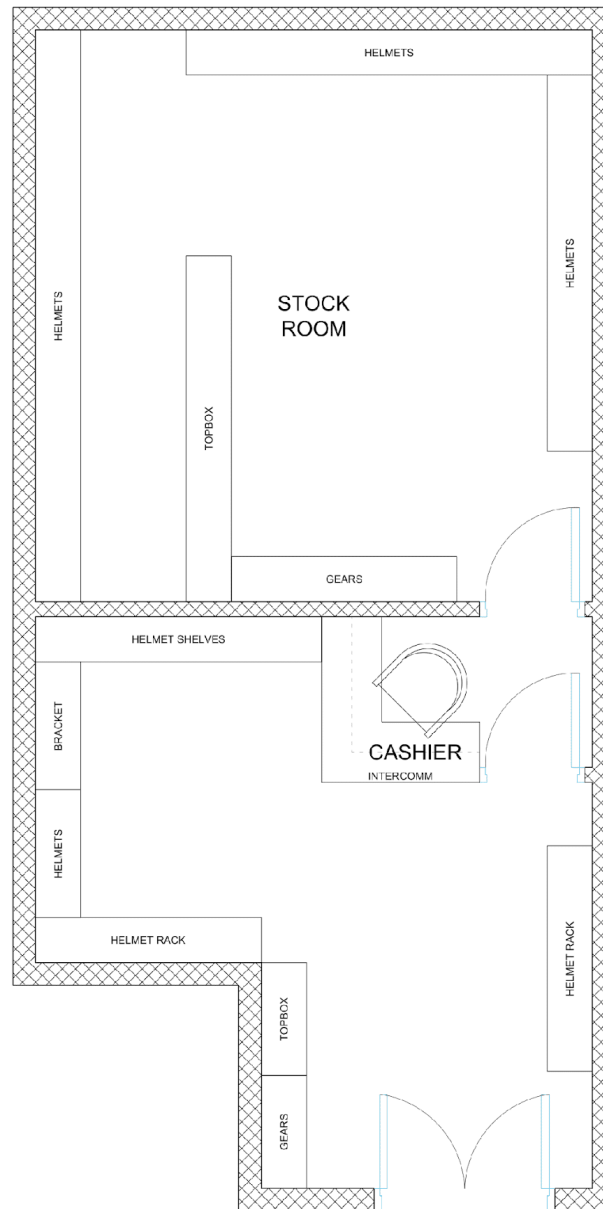


Figure 8. Office Layout

Figure 8, presents the physical arrangement of the store, highlighting key areas for product storage and customer transactions. The stock room serves as the main storage area for helmets, gears, and top boxes. Shelves and racks are strategically placed to organize products efficiently. The cashier area is positioned near the entrance, equipped with an intercom for communication. This layout ensures a systematic flow for inventory management and customer service, optimizing space for easy access and organization.

	Standard Client	Standard Web Server	Standard Application Server	Standard Database Server
Operating System	Windows	Windows / Linux	Windows / Linux	Windows / Linux
Special Software	Chrome / Microsoft Edge	XAMPP (Apache, PHP, HTML, MySQL)	- PHP - PHPMailer	phpMyAdmin
Hardware	- Intel® Core™ i3 - Quad Core CPU 4GB RAM 500GB HDD/SSD - Smartphones	- Intel® Core™ i5 - Multi-Core CPU 16GB RAM 1TB SSD	- Intel® Core™ i5 - Multi-Core CPU 16GB RAM 1TB SSD	- Intel® Core™ i5 - Multi-Core CPU 16GB RAM 1TB SSD
Network	LAN / Wi-Fi / Cellular Data	- Dual 100 Mbps Ethernet - Secure VPN	Dual 100 Mbps Ethernet	Dual 100 Mbps Ethernet

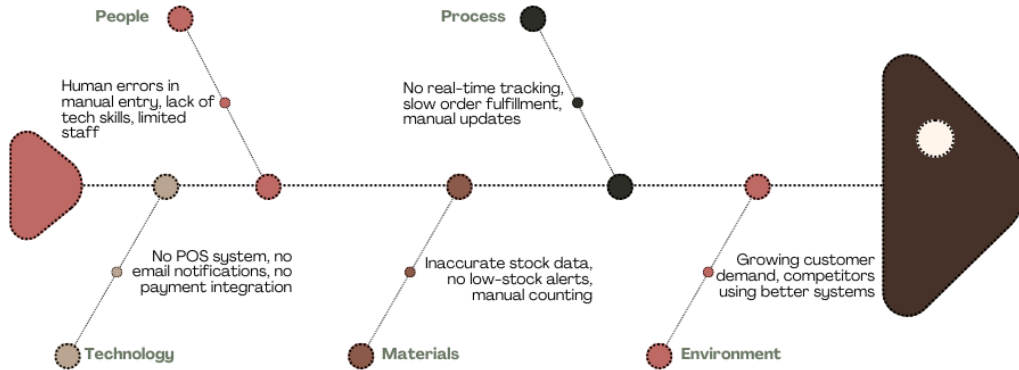
Figure 9. Hardware and Software Specifications

Figure 9, outlines the required technical setup for the system. It details the specifications for different components, including the Standard Client, Web Server, Application Server, and Database Server.

- Operating System: Windows or Linux, depending on the server type.
- Special Software: Web browsers, XAMPP for web hosting, PHP, and phpMyAdmin for database management.
- Hardware:
 - Clients require Intel Core i3, 4GB RAM, and 500GB storage.
 - Servers use Intel Core i5, 16GB RAM, and 1TB SSD.
- Network: Secure VPN and high-speed Ethernet ensure connectivity.

This setup ensures efficient performance, security, and scalability for the system.

Fishbone Diagram



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